LESSON PLAN

A/C STRUCTURE (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	17	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Symmetric flight loads, symmetric maneuvers
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Symmetric flight loads, symmetric maneuvers
5. Specific Objectives	11.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of structural load diagrams; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary structural elements of an aircraft?
(b) Define symmetric flight in terms of load distribution.
(c) List three common symmetric maneuvers.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of symmetric flight loads • Load distribution in wings and fuselage • Structural response to symmetric maneuvers • Stress analysis in symmetric conditions • Comparison with asymmetric loading effects	L & Ds	PPT, IB	15 min
(b)	• Analysis of symmetric maneuvers • G-force effects on structural integrity • Fatigue considerations in symmetric flight • Design margins for symmetric loads • Impact of maneuver intensity on structure	L & Ds	PPT, PDF	20 min
(c)	• Real-world examples of symmetric loading • Maintenance implications of symmetric stress • Future trends in symmetric load design • Safety protocols for symmetric maneuvers • Integration with other structural systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how symmetric maneuvers affect wing stress distribution.
(b) Describe the role of structural elements in handling symmetric loads.
(c) Identify potential risks associated with improper symmetric maneuver execution.

12. Home Assignment and Follow Up	Prepare a brief report on how symmetric flight loads influence the design of future aircraft structures, referencing LP 02 concepts.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Aerodynamic loading distribution, span wise and chord wise distribution, effects of sweep and distribution.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor Name, Instructor, P No.	Senior Instructor, Senior Instructor, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min